PROBLEM STATEMENT

Air pollution has become one of the major environmental and public health challenges affecting cities across India. Monitoring Air Quality Index (AQI) data manually from multiple sources makes it difficult to identify pollution trends, compare regional air quality, and generate meaningful insights for decision-making.

- There is a need for an interactive and data-driven solution that can:
- Monitor real-time AQI levels efficiently
- Identify highly polluted cities and states
- Analyze dominant pollutants affecting air quality
- Track AQI trends over time
- Present complex environmental data in a simple and visually understandable format

This project aims to develop a **Real-Time Air Quality Index (AQI) Analysis Dashboard** using **Power BI** to provide interactive visualizations, trend analysis, and pollution insights that support environmental monitoring and awareness.

Objective of the Project

- To analyze real-time AQI data across different regions
- To visualize pollution levels using interactive dashboards
- To identify major pollutants contributing to poor air quality
- To compare AQI performance between cities and states
- To support data-driven environmental analysis and reporting

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Proposed Solution

The solution is an interactive **Power BI dashboard** that integrates AQI datasets and transforms raw environmental data into meaningful visual insights through:

- KPI indicators
- Interactive maps
- Trend charts
- Pollutant analysis
- Dynamic filtering and drill-down analysis

The dashboard enables users to monitor air quality conditions effectively and gain insights into pollution patterns for better environmental understanding.